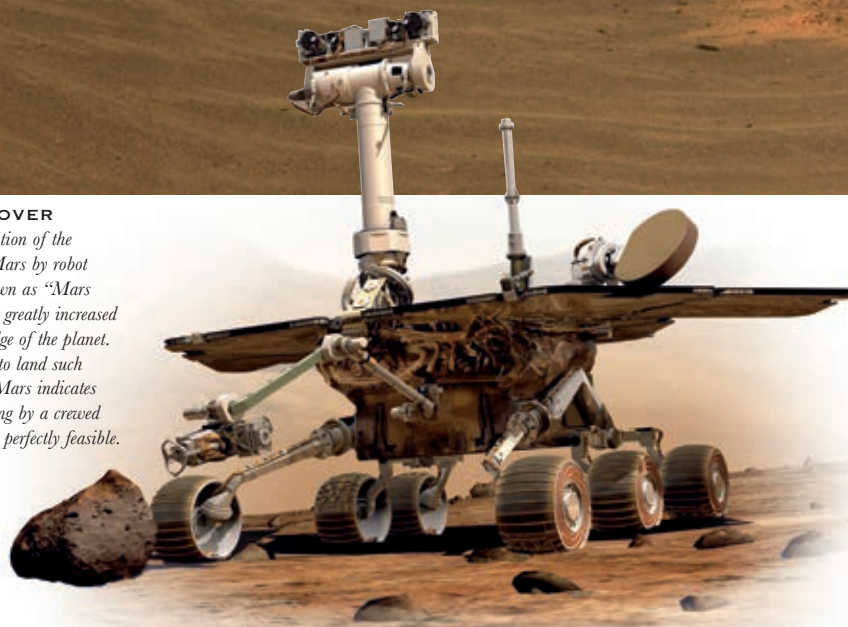




MARS ROVER

The exploration of the surface of Mars by robot vehicles known as “Mars Rovers” has greatly increased our knowledge of the planet. The ability to land such vehicles on Mars indicates that a landing by a crewed spacecraft is perfectly feasible.



recognized as posing problems far beyond any so far tackled. In principle, a human journey to the planet's surface is certainly possible – the first soft landing on Mars was achieved as long ago as 1976. However, human survival in deep space presents logistical, physical, and psychological challenges that would have to be overcome. NASA was working to a target of leaving a human footprint on Mars at some time in the 2030s, although a permanent base on the planet was expected to take much longer to establish, should it even prove achievable.

Chinese competitors

Given the past record of space exploration, there was still room for scepticism as to whether the will to fulfil this “new vision” can be sustained through decades ahead. What was needed to boost the impulse for state-sponsored human space exploration was a space race. With the demise of the Soviet Union, who would provide the competition against which the Americans would race to reach Mars? The only possible candidates were the Chinese. In October 2003 China became the third country to put a man in space when 38-year-old fighter pilot Yang Liwei orbited the globe 14 times aboard Shenzhou 5 – an evolved version of the Russian Soyuz spacecraft. Of

course, the Chinese were far behind NASA in technological development, but they had the nationalist motivation to devote major resources to space exploration. The Chinese National Space Administration (CNSA) had as part of its publicly stated mission to “become a world leader in human space exploration.” China in the early 21st century, with its vast resources generated by a rapidly expanding economy at the disposal of a powerful and combative state system, had the potential to surprise everyone with spectacular space

MARS ORBITER

The Mars Reconnaissance Orbiter, which began circling Mars in March 2006, had among its objectives the identification of landing sites for missions to the planet's surface. It was still in Mars orbit a decade after its entry into operation.

SHIELD AND CRATER ON MARS

Mars Rover Opportunity, which landed on the planet in January 2004, transmitted this image. It shows the Rover's shattered heat shield, jettisoned during descent, and the crater that the shield made on impact with the red dirt of the planet's surface. The featureless landscape suggests the cheerless prospect facing any long-term human settlers.

ventures in the future. The day that the first Chinese astronaut sets foot on the Moon, the United States will undoubtedly rediscover the motivation to back human space flight in earnest.

Space goes private

At the start of the third millennium it was still hard to see space travel to the Moon or Mars ever affecting most people's lives, except in science-fiction scenarios of escape from a devastated Earth after some natural or manmade cataclysm. In the first 50 years of state-sponsored space travel, only about 500 people had gone outside the Earth's atmosphere. The possibility of large numbers of people experiencing space travel can only become capable of realization when human space flight begins to open up to private entrepreneurship.

